

Martin Ševčík

Computer Science Student

Brno, Czech Republic
☎ +420 605 724 041
✉ sevcikm2514@gmail.com
🌐 eightl.github.io
🐙 EightL
in martin-sevcik-ba54942a3

Profile

I'm a third-year Computer Science student at Brno University of Technology (FIT VUT), continuing to Master's AI department. I have practice in Python, JavaScript/TypeScript, C++ due to projects. Interested in AI/RL and application development. I'm fond of running, reading and playing guitar in my spare time.

Education

- 2023 – present **B.Sc. in Information Technology, 3rd year**, Faculty of Information Technology, Brno University of Technology, Brno
- 2019 – 2023 **General Gymnasium**, Gymnázium Uherské Hradiště, Uherské Hradiště

Work Experience

- 2022 **Quality Control Technician (part-time)**, KOVOKON Popovice s.r.o., Kunovice
- Operated and supervised CMM measuring machines. Edited measuring scripts and analysed output data. Cooperated with production to identify and resolve non-conformities

Skills

Languages & Frameworks Python, TypeScript/JS, C++, Swift, Java, SQL; Svelte/SvelteKit, React Native (Expo), FastAPI, Qt, Supabase, Vercel, Git, Cypress, JAX, LaTeX, Bash, Docker, AI, MARL

Projects

- lerocards.com** Full-stack SaaS web app for structured mindmap flashcards-based learning; **AI-powered**; **Stripe-Supabase-Vercel** Edge Functions; **end-to-end**; Cross-platform; **Svelte**; TypeScript
- SigDriftr** Modular **NLP pipeline** for Czech media trend analysis; **RSS**; LLM signal extraction, drift detection; **contextual bandit** (LinUCB) feed selection; **FastAPI**; **sentence-transformers**.
- Interpres** Mobile app for LLM document translation, text-to-speech reading and unique notetaking; Google Cloud API (TTS, **LLM** integration, Vision **OCR**); **React Native** (Expo); **Flask**.
- Fish Rework & Baby Pets** Minecraft **datapacks/plugins**; redesigned RPG-like fishing mechanics and pet system; **Java**/datapack JSON, complex game logic; **PaperAPI**; Discord community; Custom Wiki.

Selected School Projects

- Bachelor's Thesis Research** *Agent behaviour analysis in **MARL***; Python, **JAX** environment (extreme *optimizations*: **130k+ FPS**, 328M frames); Multi-Agent **PPO**; quantitative metrics (spatial, coordination, balance) and correlation analysis; parameter-sharing and vision-radius ablations; **W&B**.
- Compiler** Zig-like; **C**; hand-written lexer and LL parser, semantic analysis, intermediate-code generator.
- Interpreter** Smalltalk-like VM with dynamic dispatch and closures; **XML parser**; Python, **PHP**; OOP.
- FSM Editor** Visual finite-state-machine editor and real-time monitor; **C++**, **Qt 5**, JSON/UDP. 2-team.
- Aquapark Simulations** Discrete-event simulation of an aquapark; queue modelling, attraction capacity, throughput statistics; **Petri** nets; **SIMLIB/C++**; 2-team.
- Love4Music** iOS App; manage favourite albums via **Spotify API**; **Swift**, SwiftUI, **SwiftData**.
- Network Tools** **C++**; **L4** port **Scanner** (TCP/UDP, IPv4/IPv6), CLI **Chat Client** (custom TCP/UDP protocol); **SNMP-to-OpenTelemetry** bridge (gRPC/OTLP).

Language Skills – Czech (native), English (B2)